

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SP8 - General Insurance: Pricing

Total Marks: 100

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1)

a) The products to be suggested to Mumbai Indians are-

(i) **Personal Accident Insurance**

Injury of any player while practice may lead to absenteeism in rival matches, leading to probable financial loss to the team

(ii) **Fidelity Guarantee**

Any treachery or malicious involvement of players may lead to temporary ban or financial liability to the team

(iii) **Motor Insurance**

Team owns buses and other vehicles for commercial use. Property Damage to the vehicle or Third-Party Liability resultant of the vehicle may lead to financial liability for the team

(iv) **Property Damage Insurance**

The team may own practice areas/ offices / changing rooms / grounds / pitches etc. for practicing and other administrative work. Any loss due to fire or allied perils may lead to financial loss to the team

(v) **Public Liability Insurance**

An example of Public Liability claim arising to the team may be ball hit by a team player hitting a passer-by causing physical injury or death.

(vi) **Professional Indemnity Insurance**

Besides players, a team consists of many professionals supporting them like Doctors, Coaches, Physiotherapists, etc. Any professional negligence on their part may lead to financial liability for the team

(vii) **Group Mediclaim Insurance**

Like Personal Accident, any sickness or hospitalization may lead to absenteeism of players in important matches, leading to probable financial loss to the team

(viii) **Event Cancellation**

The team is liable to make payment to players irrespective of the whether or not matches are played. However, in case due to weather conditions or other unavoidable circumstances the matches get post-poned or cancelled, the team may face financial losses.

(8)

b) Information required from the team manager may include-

- Number of players and other type of professionals
- Level and nature of financial agreement with each of them
- Personal information like age, health status etc.
- Types of insurance covers to be purchased
- Value of assets
- Nature of assets & age
- List of precautionary measures taken for lives and assets

(3)

c) Data sources-

- Past medical history for each team player from the team manager
- Newspaper articles on game records for each player
- Internal experience from other sport players highlighting difference in risk due to healthier lifestyle
- Reinsurer's expertise on underwriting other cricket matches

(2)

d) Cost of each layer will be calculated as follows:

5 Cr xs 2 Cr = 2 Cr to 7 Cr = $1.6 + 0.5 + 0.8 = 2.9$ Cr

1 free reinstatement = 7 Cr to 12 Cr = Free = 0

1st paid reinstatement = 12 Cr to 17 Cr = 1 Cr

2nd paid reinstatement = 17 Cr to 22 Cr = $0.4 + 0.13 = 0.53$ 0.5 marks]

Total cost = $2.9 + 1 + 0.53 = 4.43$ Cr

(3)
[16]

Q. 2)

- a) A unit of measure determining how much risk there is within a policy is called its Measure of Exposure

Yield per hectare or Yield per square foot is an ideal measure of exposure for Crop Insurance as-

- It is measurable
 - Reflects varying yield by type of crop
 - Higher the area, means higher the value of yield and hence higher the probable loss
- (3)

b)

- The Company will need to consider whether there is **enough credible data** for each crop grown in each district across each season
 - **Adjustment of past data** for the volatility observed in weather trends each year
 - **A detailed pricing model** will need to be constructed and validations around the following will need to be made
 - i. Model Error
 - ii. Parameter Error
 - iii. Random or Process Error
 - Appropriate **adjustments for market situations** will need to be made like changing demand pattern for each crop, change in levels of Minimum Selling Price (MSP) set by the Government etc.
 - **Impact on capital requirements and solvency** for writing a new line of business based on prevailing regulatory norms
 - **Other non-pricing impacts** like regulatory requirements, cost synergies, competitive advantage etc.
- (4)

c)

- The form of reinsurance best suited for the insurance company would depend upon factors like-
 - Current level of solvency
 - Diversification benefits available
 - Risk appetite of the shareholders
 - Technical expertise of pricing and claims settlement available internally
 - Proportional reinsurance may only be suited for small insurers with-
 - Lack of pricing expertise
 - Requiring share of small claims too
 - However, based on risk of accumulation involved due to weather conditions affecting various adjoining farms at the same time and sales being done via district allocation by the Government, XoL or Stop Loss reinsurance is most suited.
 - The Company may decide to mutually agree to a customised cover involving various layers of nested covers for example – no cover upto 10 cr, 20% cover from 10-25 Cr and full cover upto 50 Cr; but only if Loss Ratio remains below 200%, further no cover above 200% of Loss Ratio
 - The Company may top it up with another Cat XOL cover with a defined 8-10 Cr Hours clause for any accumulated period of heavy rain or drought
- (6)

d)

- Level of credibility depends upon the inherent volatility in the available claims data
- For Frequency-Severity approach, credibility needs to be validated separately for claim count and claim amount data
- Crop Insurance is a low frequency-high severity event, compared to Motor Own Damage which is high frequency-low severity in nature
- Credibility depends upon the volume of base for a metric i.e. exposure for frequency & claim count for severity
- So, if across 15 years only 2 claims have occurred in Crop insurance, it may suggest a low frequency but may be a credible phenomenon for frequency as such losses are not expected to occur every year but may not be credible for severity calculations.

(4)

e)

- Claims distributions are typically positively skewed in nature i.e. many small claims and few large claims are expected to occur
- So level of volatility in attritional claims will be lower than than of catastrophe claims
- Hence large data volumes may provide sufficient credibility to attritional claims but may still not be enough for catastrophe claims as they have not have occurred even once in the data available. For example, Earthquakes have a return period of 1 in 250 years
- Hence catastrophe modelling follows a different and more complex method involving studies of seismology, topology, etc. along with past occurrence data
- Sometime due to lack of history of such events, dummy events are required to be created in the data with a expected impact study; followed by thousands of simulations being run to cover various plausible scenarios of occurrence of such events.

(4)

[21]

Q. 3)

a) Reasons of not running television advertisements-

- Such levels of marketing expenses are not included in current levels of pricing
- Doesn't align with Company's marketing strategy & brand image
- Product positioning is different than competitor and doesn't require television advertisements
- Target cohorts have higher burn cost than competitors like your Company may be targeting more of Commercial Vehicle business vs. higher Private Car business of competitor
- Pricing is dated and needs to undergo changes
- Required levels of prices to run these advertisements will fall outside regulatory approved ranges

b) Reasons of difference in premium quoted by competitor in advertisement-

- Difference quoted is gross of GST, hence quantum is higher
- Difference quotes is only for a specific cohort which may be loss making or outside target market for your company
- Business Acquisition Expenses may differ as theirs is only a digital company, whereas yours is dominated by traditional channels of business like agency
- Pricing strategy may be different as they are newer in the market and may be writing business at higher loss ratios
- Reinsurance support available may be more expensive for your Company

[8]

Q. 4)**a) Details of work of a Pricing Actuary may include-**

- Contribution towards product development by confirming whether or not designed coverage can be priced
 - i. Is required data available?
 - ii. Is risk measurable objectively?
 - iii. Is liability limited?
 - iv. Does price fall within regulatory requirements?
 - v. Is price marketable i.e. range bound with competitors and logically lower than sum insured
- Rating Analyses of products
 - (a) Estimate cost of incurred claims in recent periods
 - (b) Estimate levels of profitability at current premium rates
 - (c) Project claims over period where new rates will be charged and its corresponding claims will emerge
 - (d) Review suitability of current rating structure and consider possible changes
 - (e) Market test final rates with competitors
 - (f) Apply adjustment for lifetime value
 - (g) Analyse profitability of older years basis new rates
- Expense Analyses
 - (a) Distribute current levels of expenses into short term, long term, fixed & variable, direct vs indirect expense heads
 - (b) Study current levels of persistency and check expense trends with respect to change in persistency
 - (c) Allocating Company level expenses to each product portfolio scientifically
 - (d) Evaluating sensitivity of allocation methodology on level of overall profitability
- Portfolio monitoring
 - (a) Assess emerging experience against expected experience
 - (b) Managing risk accumulations for purchase of appropriate reinsurance cover
 - (c) Gaining market intelligence around strategy of competitors and changing underwriting cycle
 - (d) Meeting regulatory and reinsurance data submission requirements
 - (e) Disclosure to various stakeholders for performance review
 - (f) Assist in reserving inputs
- Other tasks:
 - (a) Carry out profit testing exercises
 - (b) Estimate price elasticity curves
 - (c) Create lifetime value models
 - (d) Redesign rating tariffs
 - (e) Assist in reinsurance decisions
 - (f) Capital and catastrophe modelling etc.

(10)

b) Reasons of failing prediction of a lottery

- Past data may not be easily available to predict future outcomes
- Aggregate models predict combined outcome of occurrence and impact i.e. total amount won or lost in each round in the past, whereas requirement would to only predict the occurrence separately
- Model may be subject simulation error as the outcome would share range of values whereas requirement would be a point estimate

- Running a stochastic model will required high levels of computational power
- It assumes each round of lottery will be independent which may be controlled by the casino house
- It would not consider initial buy in cost of the lottery

(4)

[14]**Q. 5) a)**

- Motor Third Party is a long-tailed business or has a large IBNR component or takes many years to settle claims.
- Further, the claims frequency is low, so a greater number of claims are required to achieve full credibility.
- Atleast 7-8 years of internal data should be used for pricing.

(2)

b)

- Pricing should be done basis Accident Year data only
- In reporting year, as newer claims get reported, value of the cohort changes.
- Claim trending is easier in Accident Year than in Reporting Year

(2)

c) The regulator has been announcing price changes in many years. So Loss Ratio experience is each year is not comparable unless its adjusted to same price levels. Hence, its better to adjust all year's premium to current levels of price.

(1)

d)

- Companies holds statistical claim reserves to avoid volatility in its financial statement by reporting or closure of one large claim
- Companies hold case reserves to help claim managers to understand true liability against the claim and helps them validate this against information available
- For the purpose of pricing, true liability needs to be established against the premium charged for maintaining consistency and comparing performance against years. Hence, case reserves should be used for pricing

(3)

e) For the purpose of pricing, fully developed claims information is to be used. So incurred claims should be adjusted for undeveloped liability before using for pricing.

(Reserving via Chain-Ladder method requires earliest year to be fully developed and doesn't affect pricing)

(1)

f)

- Pricing requires risk cohorts to be homogenous in nature
- Reduction of Commercial Vehicle (CV) business is expected to create a major difference in risk profile of business.
- Pricing team can either exclude the earlier years of business from their data or remove impact of CV business only from both premium and claims.
- Considering that by removal of impact of CV business, higher number of years of business can be used for pricing i.e. higher chances of achieving credibility, that method should be opted.

(4)

g) Premium & Claim amounts in Cr.

AY	Premium	Paid Claims	Case Reserve	Incurred Claims = Paid Claims + Case reserves
2016-17	1,649	825	577	1,402
2017-18	1,929	1,447	386	1,833
2018-19	2,238	694	560	1,253
2019-20	2,350	635	705	1,340
2020-21	3,032	940	1,061	2,001
2021-22	3,759	827	1,504	2,331
2022-23	4,210	800	1,895	2,695
2023-24	5,347	535	2,673	3,208

AY	Incurred Claims	Factor	Ultimate Claim Amt = Incurred claims / (1- Factor)
2016-17	1,402	13.0%	1,611
2017-18	1,833	14.5%	2,144
2018-19	1,253	17.5%	1,519
2019-20	1,340	22.0%	1,717
2020-21	2,001	27.0%	2,741
2021-22	2,331	29.0%	3,283
2022-23	2,695	33.0%	4,022
2023-24	3,208	35.0%	4,936

AY	Premium	Ultimate Claim Amt (ULC)	CV Premium %	CV ULR%	Ultimate Claim Amt excl. CV = ULC – (Premium * CV Premium% * CV ULR%)
2016-17	1,649	1,611	20%	110%	1,248
2017-18	1,929	2,144	20%	110%	1,719
2018-19	2,238	1,519	20%	110%	1,027
2019-20	2,350	1,717	20%	110%	1,200
2020-21	3,032	2,741	20%	110%	2,074
2021-22	3,759	3,283	10%	110%	2,869
2022-23	4,210	4,022	10%	110%	3,559
2023-24	5,347	4,936	10%	110%	4,347

(12)

- h) Total Ultimate Claims from 2016-24 = 18,044
 Total Exposure from 2016-24 = 122,136
 Average Claim per unit of exposure from 2016-24 = 0.15
 Targeted LR = 65%
 Required premium per unit of exposure = $0.15 / 65\% = 0.227$
 Premium rate in 2023-24 = Premium / Exposure = $5,347 / 25,547 = 0.209$
 Hence, rate increase = Required Premium / Premium rate in 2023-24 - 1 = $0.227 / 0.209 - 1 = 8.59\%$

(4)

- i) If everything is equal, Average claim per unit of exposure would continue to be 0.15 for the

next 3 years

[1 mark]

With incremental 2% premium increase in next 3 years, the premium rates would be

AY	Premium Rate	Loss Ratio
2024-25	0.213	69%
2025-26	0.218	68%
2026-27	0.222	67%

As the required premium rate from part h) above is 0.227 and a consecutive 2% increase over a period of 3 years would only fetch 0.222; resulting a Loss Ratio higher than 65% even after 3 years

Hence, an alternative scenario would be where average Loss Ratio of the 3 years would be 65%, i.e. 2023-24 would be slightly above 65%, 2024-25 would be 65% and 2025-26 would slightly below 65%

Using standard interpolation techniques, the year on year increase required to achieve this is 4.5%

(12)
[41 Marks]
